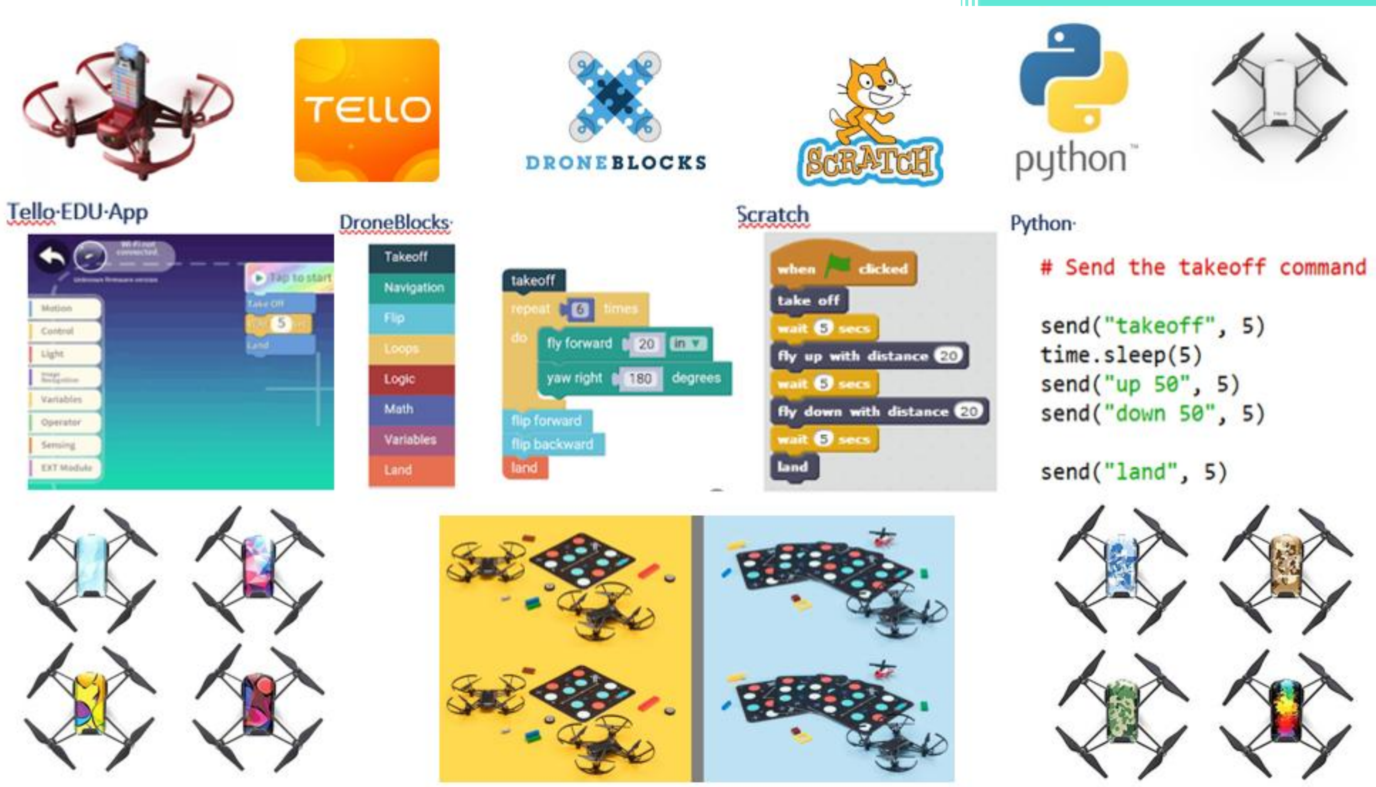
2023



Tello Edu drón

Robomaster TT

Tartalom

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TVARY

101. úloha: Trojuholníkový pohyb s otáčaním

Triangle Movement with Rotations

```python
from djitellopy import Tello
import time

# Program 2: Triangle Movement with Rotations
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(3):
    tello.move_forward(50)
    time.sleep(1)
    tello.rotate_clockwise(120)
    time.sleep(1)

tello.land()
```

Uloha_101.py

102. úloha: Kruhový pohyb so zmenou výšky

Circle Movement with Altitude Changes

```
from djitellopy import Tello
import time
import math

# Program 3: Circle Movement with Altitude Changes
tello = Tello()
tello.connect()
tello.takeoff()

radius = 50
circumference = 2 * math.pi * radius

for _ in range(36):
    tello.rotate_clockwise(10)
    time.sleep(0.1)
    tello.move_forward(circumference/36)
    time.sleep(0.1)
    tello.move_up(2)
    time.sleep(0.1)

tello.land()
```

Uloha_102.py

103. úloha: Obdĺžnikový pohyb s kombinovanými operáciami

Rectangle Movement with Combined Actions

```python
from djitellopy import Tello
import time

# Program 4: Rectangle Movement with Combined Actions
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(2):
    tello.move_forward(60)
    time.sleep(1)
    tello.rotate_clockwise(90)
    time.sleep(1)
    tello.move_forward(30)
    time.sleep(1)
    tello.rotate_counter_clockwise(90)
    time.sleep(1)

tello.land()
```

Uloha_103.py

104. úloha: Kruhový pohyb so saltami

Circular Movement with Flips

```python
from djitellopy import Tello
import time
import math

# Program 9: Circular Movement with Flips
tello = Tello()
tello.connect()
tello.takeoff()

radius = 30
circumference = 2 * math.pi * radius

for _ in range(36):
    tello.rotate_clockwise(10)
    time.sleep(0.1)
    tello.move_forward(circumference/36)
    time.sleep(0.1)
    tello.flip("r")  # Flip right
    time.sleep(0.1)

tello.land()
```

Uloha_104.py

105. úloha: Štvorcový pohyb s vznášaním

Square Movement with Hover

```python
from djitellopy import Tello
import time

# Program 17: Square Movement with Hover
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(4):
    tello.move_forward(40)
    time.sleep(1)
    tello.move_left(10)
    time.sleep(1)
    tello.move_backward(40)
    time.sleep(1)
    tello.move_right(10)
    time.sleep(1)
    tello.send_rc_control(0, 0, 0, 0)  # Hover in place
    time.sleep(1)

tello.land()
```

Uloha_105.py

106. úloha: Kruhový pohyb so zmenou výšky

Circle Movement with Change in Altitude

```python
from djitellopy import Tello
import time
import math

# Program 18: Circle Movement with Change in Altitude
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(2):
    tello.move_up(20)
    time.sleep(1)
    tello.rotate_clockwise(180)
    time.sleep(1)
    tello.move_down(20)
    time.sleep(1)
    tello.rotate_clockwise(180)
    time.sleep(1)

tello.land()
```

Uloha_106.py

107. úloha: Vlastný štvorcový pohyb so vstupom od používateľa

Custom Square Movement with User Input

```python
from djitellopy import Tello

# Program 20: Custom Square Movement with User Input
tello = Tello()
tello.connect()
tello.takeoff()

print("Enter side length of the square (in cm): ")
side_length = float(input())

for _ in range(4):
    tello.move_forward(side_length)
    tello.rotate_clockwise(90)

tello.land()
```

Uloha_107.py

108. úloha: Kruhový pohyb s náhodnými obratmi

Circular Movement with Randomized Flips

```
from djitellopy import Tello
import time
import math
import random

# Program 21: Circular Movement with Randomized Flips
tello = Tello()
tello.connect()
tello.takeoff()

radius = 30
circumference = 2 * math.pi * radius

    # Move in a circular path
for _ in range(36):  # 36 steps for a smoother circle
    tello.rotate_clockwise(10)
    time.sleep(0.1)
    tello.move_forward(circumference / 36)
    time.sleep(0.1)
    flip_direction = random.choice(["l", "r", "f", "b"])
    tello.flip(flip_direction)
    time.sleep(0.1)

tello.land()
```

Uloha_108.py

109. úloha: Obdĺžnikový pohyb s meniacou sa rýchlosťou

Rectangle Movement with Variable Speeds

```python
from djitellopy import Tello
import time

# Program 23: Rectangle Movement with Variable Speeds
tello = Tello()
tello.connect()
tello.takeoff()

speeds = [30, 50, 70, 90]

for i in range(4):
    tello.move_forward(40, speeds[i % len(speeds)])
    time.sleep(2)
    tello.move_right(20, speeds[i % len(speeds)])
    time.sleep(2)

tello.land()
```

Uloha_109.py

CIKCAK

110. úloha: Stúpajúci a klesajúci cikcak pohyb

Ascending and Descending Zigzag Movement

```python
from djitellopy import Tello
import time

# Program 5: Ascending and Descending Zigzag Movement
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(4):
    tello.move_up(30)
    time.sleep(1)
    tello.move_right(20)
    time.sleep(1)
    tello.move_down(30)
    time.sleep(1)
    tello.move_right(20)
    time.sleep(1)

tello.land()
```

Uloha_110.py

111. úloha: Cikcak

Zigzag Movement

```python
from djitellopy import Tello
import time

# Program 7: Figure-Eight Movement
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(2):
    tello.move_forward(50)
    time.sleep(1)
    tello.move_left(30)
    time.sleep(1)
    tello.move_forward(50)
    time.sleep(1)
    tello.move_right(30)
    time.sleep(1)

tello.land()
```

Uloha_111.py

112. úloha: Diagonálny cikcak pohyb doprava so saltami

Diagonal Movement with Flips

```
from djitellopy import Tello
import time

# Program 11: Diagonal Movement with Flips
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(2):
    tello.move_forward(30)
    time.sleep(1)
    tello.move_right(30)
    time.sleep(1)
    tello.flip("f")  # Flip forward
    time.sleep(1)

tello.land()
```

Uloha_112.py

113. úloha: Cikcak pohyb so zmenou výšky

Zigzag Movement with Altitude Changes

```python
from djitellopy import Tello
import time

# Program 12: Zigzag Movement with Altitude Changes
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(3):
    tello.move_forward(40)
    time.sleep(1)
    tello.move_right(20)
    time.sleep(1)
    tello.move_up(20)
    time.sleep(1)

tello.land()
```

Uloha_113.py

114. úloha: Cikcak pohyb s náhodnými otočkami a saltami

Zigzag with Randomized Rotations and Flips

```python
from djitellopy import Tello
import time
import random

# Program 13: Zigzag with Randomized Rotations and Flips
tello = Tello()
tello.connect()
tello.takeoff()

# Zigzag movement with randomized rotations
for _ in range(4):
    tello.rotate_clockwise(random.randint(45, 180))
    time.sleep(1)
    tello.move_forward(random.randint(20, 40))
    time.sleep(1)

    # Perform a random flip
    flip_direction = random.choice(["l", "r", "f", "b"])
    tello.flip(flip_direction)
    time.sleep(1)

tello.land()
```

Uloha_114.py

115. úloha: Diagonálny cikcak pohyb doľava s náhodnou zmenou výšky

Hovering Square with Randomized Altitude Changes

```
from djitellopy import Tello
import time
import random

# Program 22: Hovering Square with Randomized Altitude Changes
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(4):
    tello.move_forward(30)
    time.sleep(1)
    tello.move_up(random.randint(10, 30))
    time.sleep(1)
    tello.move_left(30)
    time.sleep(1)
    tello.move_down(random.randint(10, 30))
    time.sleep(1)

tello.land()
```

Uloha_115.py

116. úloha: Diagonálny cikcak pohyb s náhodnými saltami

Diagonal Zigzag with Randomized Flips

```
from djitellopy import Tello
import time
import random

# Program 26: Diagonal Zigzag with Randomized Flips
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(2):
    tello.move_forward(30)
    time.sleep(1)
    tello.move_right(30)
    time.sleep(1)
    flip_direction = random.choice(["l", "r", "f", "b"])
    tello.flip(flip_direction)
    time.sleep(1)

tello.land()
```

Uloha_116.py

117. úloha: Cikcak pohyb doľava s rýchlosťou určenou používateľom

Square Movement with User-Defined Speed

```
from djitellopy import Tello
import time

# Program 27: Square Movement with User-Defined Speed
tello = Tello()
tello.connect()
tello.takeoff()

speed = int(input("Enter speed (20-100): "))

for _ in range(4):
    tello.move_forward(40, speed)
    time.sleep(2)
    tello.move_left(20, speed)
    time.sleep(2)

tello.land()
```

[Uloha_117.py]

118. úloha: Diagonálny cikcak doľava s náhodnými saltami

Diagonal Square with Randomized Flips

```
from djitellopy import Tello
import time
import random

# Program 30: Diagonal Square with Randomized Flips
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(2):
    tello.move_forward(30)
    time.sleep(1)
    tello.move_left(30)
    time.sleep(1)
    flip_direction = random.choice(["l", "r", "f", "b"])
    tello.flip(flip_direction)
    time.sleep(1)

tello.land()
```

[Uloha_118.py]

ĎALŠIE POHYBY

119. úloha: Pohyb dopredu doľava so saltom

Forward Movement with Flips

```
from djitellopy import Tello
import time

# Program 1: Square Movement with Flips
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(4):
    tello.move_forward(50)
    time.sleep(1)
    tello.flip("l")  # Flip left
    time.sleep(1)

tello.land()
```

Uloha_119.py

120. úloha: Špirálový pohyb nahor

Spiraling Upward Movement

```python
from djitellopy import Tello
import time
import math

# Program 6: Spiraling Upward Movement
tello = Tello()
tello.connect()
tello.takeoff()

for i in range(20):
    tello.move_up(10)
    time.sleep(1)
    tello.rotate_clockwise(360/20)
    time.sleep(1)

tello.land()
```

[Uloha_120.py]

121. úloha: Schodovitý pohyb bez otočiek

Staircase Movement with Altitude Changes

```python
from djitellopy import Tello
import time

# Program 8: Square Movement with Altitude Changes
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(4):
    tello.move_forward(50)
    time.sleep(1)
    tello.move_up(20)
    time.sleep(1)

tello.land()
```

Uloha_121.py

122. úloha: Pohyb v tvare diamantu

Diamond Pattern Movement

```
from djitellopy import Tello
import time

# Program 10: Diamond Pattern Movement
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(4):
    tello.move_forward(50)
    time.sleep(1)
    tello.rotate_clockwise(45)
    time.sleep(1)

tello.land()
```

Uloha_122.py

123. úloha: Náhodná sekvencia pohybov

Random Movement Sequence

```python
from djitellopy import Tello
import time
import random

# Program 14: Random Movement Sequence
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(5):
    move_type = random.choice(["forward", "right", "left", "back"])
    move_distance = random.randint(20, 50)

    if move_type == "forward":
        tello.move_forward(move_distance)
    elif move_type == "right":
        tello.move_right(move_distance)
    elif move_type == "left":
        tello.move_left(move_distance)
    elif move_type == "back":
        tello.move_back(move_distance)

    time.sleep(1)

tello.land()
```

[Uloha_123.py]()

124. úloha: Pohyb hore a dole s otáčaním

Up and Down Movement with Spin

```
from djitellopy import Tello
import time

# Program 15: Up and Down Movement with Spin
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(3):
    tello.move_up(30)
    time.sleep(1)
    tello.rotate_counter_clockwise(360)
    time.sleep(1)
    tello.move_down(30)
    time.sleep(1)

tello.land()
```

Uloha_124.py

125. úloha: Schodovitý pohyb so saltami

Staircase Pattern with Flips

```
from djitellopy import Tello
import time

# Program 16: Staircase Pattern with Flips
tello = Tello()
tello.connect()
tello.takeoff()

for step in range(1, 6):
    tello.move_forward(20)
    time.sleep(1)
    tello.move_up(step * 10)
    time.sleep(1)
    tello.flip("r")  # Flip right
    time.sleep(1)

tello.land()
```

Uloha_125.py

126. úloha: Pohyb dopredu a dozadu so zmenou rýchlosti

Forward and Backward Movement with Speed Variation

```python
from djitellopy import Tello
import time

# Program 19: Forward and Backward Movement with Speed Variation
tello = Tello()
tello.connect()
tello.takeoff()

for speed in range(20, 61, 20):
    tello.move_forward(50, speed)
    time.sleep(2)
    tello.move_backward(50, speed)
    time.sleep(2)

tello.land()
```

Uloha_126.py

127. úloha: Stúpajúca špirála s náhodným otáčaním

Ascending Spiral with Randomized Rotations

```
from djitellopy import Tello
import time
import random

# Program 24: Ascending Spiral with Randomized Rotations
tello = Tello()
tello.connect()
tello.takeoff()

for i in range(20):
    tello.move_up(10)
    time.sleep(1)
    tello.rotate_clockwise(random.randint(10, 30))
    time.sleep(1)

tello.land()
```

Uloha_127.py

128. úloha: Štvorcový pohyb s nepretržitým otáčaním

Square Movement with Continuous Spin

```
from djitellopy import Tello
import time

# Program 25: Square Movement with Continuous Spin
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(4):
    tello.move_forward(40)
    time.sleep(1)
    tello.rotate_clockwise(360, speed=50)
    time.sleep(1)

tello.land()
```

Uloha_128.py

129. úloha: Pohyb dopredu s nepretržitou zmenou výšky

Square Movement with Continuous Altitude Change

```
from djitellopy import Tello
import time

# Program 28: Square Movement with Continuous Altitude Change
tello = Tello()
tello.connect()
tello.takeoff()

for _ in range(4):
    tello.move_forward(40)
    time.sleep(1)
    tello.move_up(10)
    time.sleep(1)
    tello.move_down(10)
    time.sleep(1)

tello.land()
```

Uloha_129.py

130. úloha: Stúpajúce a klesajúce schody

Ascending and Descending Staircase

```python
from djitellopy import Tello
import time

# Program 29: Ascending and Descending Staircase (Corrected)
tello = Tello()
tello.connect()
tello.takeoff()

# Ascending
for step in range(1, 6):
    tello.move_forward(20)
    time.sleep(1)
    tello.move_up(step * 10)
    time.sleep(1)

# Descending
for step in range(5, 0, -1):
    tello.move_forward(20)
    time.sleep(1)
    tello.move_down(step * 10)
    time.sleep(1)

tello.land()
```

Uloha_130.py

MISSIONPAD

131. úloha: Pohyb sledujúci MissionPad a nahrávanie videa 1

MissionPad Program 1

```python
from djitellopy import Tello
import cv2
import threading

# Set the speed
speed = 50

# Initialize the Tello drone
drone = Tello()
drone.connect()
drone.streamon()
drone.enable_mission_pads()

# Takeoff
drone.takeoff()
# Command execution function
def execute_commands():
    drone.move_down(20)
    drone.go_xyz_speed_mid(100, 0, 50, speed, 1)
    drone.go_xyz_speed_mid(0, 70, 70, speed, 2)
    drone.go_xyz_speed_mid(-100, 0, 50, speed, 3)
    drone.go_xyz_speed_mid(0, -70, 70, speed, 4)
    drone.land()

# Video display function
def display_video():
    while True:
        frame = drone.get_frame_read().frame
        frame = cv2.resize(frame,(1600,1300))
        cv2.imshow("Output", frame)
        if cv2.waitKey(1) & 0xFF == ord("q"):
            drone.land()
            cv2.destroyAllWindows()
            break

# Create and start the threads
command_thread = threading.Thread(target=execute_commands)
video_thread = threading.Thread(target=display_video)

command_thread.start()
video_thread.start()

# Wait for both threads to finish
command_thread.join()
video_thread.join()
```

Uloha_131.py

132. úloha: Pohyb sledujúci MissionPad a nahrávanie videa 2

MissionPad Program 2

```python
from djitellopy import Tello
import cv2
import threading

# Set the speed
speed = 50

# Initialize the Tello drone
drone = Tello()
drone.connect()
drone.streamon()
drone.enable_mission_pads()

# Takeoff
drone.takeoff()
# Command execution function
def execute_commands():
    drone.go_xyz_speed_mid(50, 35, 100, speed, 1)
    drone.go_xyz_speed_mid(50, 35, 50, speed, 5)
    drone.go_xyz_speed_mid(-50, -35, 100, speed, 3)
    drone.go_xyz_speed_mid(-50, 35, 50, speed, 5)
    drone.go_xyz_speed_mid(50, -35, 100, speed, 4)
    drone.go_xyz_speed_mid(50, -35, 50, speed, 5)
    drone.go_xyz_speed_mid(-50, 35, 100, speed, 2)
    drone.land()

# Video display function
def display_video():
    while True:
        frame = drone.get_frame_read().frame
        frame = cv2.resize(frame,(1600,1300))
        cv2.imshow("Output", frame)
        if cv2.waitKey(1) & 0xFF == ord("q"):
            drone.land()
            cv2.destroyAllWindows()
            break

# Create and start the threads
command_thread = threading.Thread(target=execute_commands)
video_thread = threading.Thread(target=display_video)

command_thread.start()
video_thread.start()

# Wait for both threads to finish
command_thread.join()
video_thread.join()
```

Uloha_132.py